



Hardware Requirements

Recommended machine specifications for running Prismo operator nodes.

Sized for current testnet load (~100 TPS peak, low utilization). **Re-evaluate before mainnet** — expect higher sustained throughput, larger state, and stricter latency SLAs. The numbers below are the floor for testnet, an opening bid for mainnet.

■ Per Node Role

Role	vCPU	RAM	Disk	Disk type	Network	Notes
RPC Node	8	32 GB	1 TB	NVMe SSD	100 Mbps	Heavier with archive mode
Full Node (state-verifying)	8	32 GB	1 TB	NVMe SSD	100 Mbps	Same as RPC + L1 archive access
Watchtower	2	4 GB	50 GB	SSD	25 Mbps	Plus query access to a full node
Bridge Indexer	4	8 GB	200 GB	SSD	25 Mbps	Postgres on same host or external

For mainnet, double RAM as a starting assumption and plan for at least 2 TB of disk per node — the L1 fee market will keep state growth higher than testnet.

■ Disk Growth

Estimated growth per month at testnet load:

Role	Pruned	Archive
RPC Node	~30 GB	~150 GB
Full Node	~30 GB	~150 GB
Bridge Indexer	~5 GB	n/a

Pruned mode keeps recent state + headers. Archive keeps full historical state — required if you want to serve `debug_traceTransaction` for old blocks. Mainnet figures will be revised once load data is in.

■ L1 Dependency

Every L2 node needs an L1 RPC endpoint on the network it follows.

Network	L1	Self-host option	Public-RPC option
Testnet	Sepolia	geth/reth + lighthouse/prysm, ~1 TB	Alchemy / Infura / Ankr / dRPC
Mainnet	Ethereum	geth/reth + lighthouse/prysm, ~2.5 TB	Alchemy / Infura / Ankr / dRPC

Recommendation:

- **Self-host for full nodes and watchtowers** — they hammer L1 during catch-up; public providers will rate-limit.
- **Public RPC is fine for an RPC node** that consumes the sequencer data stream and only does light L1 reads.
- **Run inside same compose / k8s** as the L2 node. The compose template selects `--sepolia` or `--mainnet` from `L1_NETWORK_FLAG` in the active network env.

■ Cloud Sizing Reference

Provider	RPC node	Full node	Watchtower	Bridge indexer
AWS	<code>m6i.2xlarge</code> + 1 TB gp3	<code>m6i.2xlarge</code> + 1 TB gp3	<code>t3.medium</code> + 50 GB gp3	<code>t3.large</code> + 200 GB gp3
Hetzner	<code>CCX23</code> + 960 GB NVMe	<code>CCX23</code> + 960 GB NVMe	<code>CX22</code> + 80 GB	<code>CCX13</code> + 240 GB
GCP	<code>n2-standard-8</code> + 1 TB pd-ssd	<code>n2-standard-8</code> + 1 TB pd-ssd	<code>e2-medium</code> + 50 GB pd-ssd	<code>e2-standard-2</code> + 200 GB pd-ssd

Monthly cost rough order (testnet sizing): AWS ~\$280, Hetzner ~\$45, GCP ~\$300 for the RPC tier.

■ Latency Considerations

- Sequencer data stream is in `us-east-1` (testnet; mainnet location TBD). RPC nodes in Asia/EU lag ~150–250 ms behind sequencer head — fine for normal use.
- Watchtowers can run anywhere — they only read L1 events.
- Bridge indexer is read-mostly; latency unimportant.